SID 学号：　　　　　　　　　　　　NAME 姓名：

Homework 9

✧ Sine-wave oscillator：P471：8.6/8.7/8.8/8.10/8.11/8.12

8.6 电路如**图 P8.6** 所示，试求解：(1)R_W 的下限值；(2)振荡频率的调节范围。

8.6 An oscillating circuit is shown in figure p8.6, R_2 can be adjusted from 0 to 100kΩ, ask:

(1) The lower limit value of R_W to oscillate;

(2) The range of oscillation frequency.

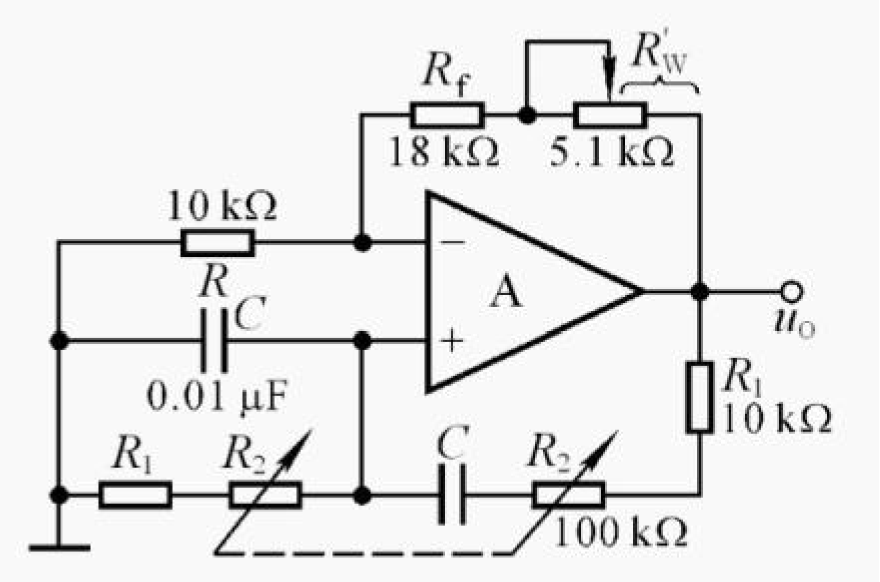

图 P8.6

8.7 电路如**图 P8.7** 所示，稳压管起稳幅作用，其稳压值为±6V。试估算：

(1)输出电压不失真情况下的最大有效值；

(2)振荡频率。

8.7 An oscillating circuit is shown in figure p8.7. The working voltage of voltage regulator D_Z is ±6V. Please estimate:

(1) The maximum effective value of the output voltage without distortion;

(2) The oscillation frequency.

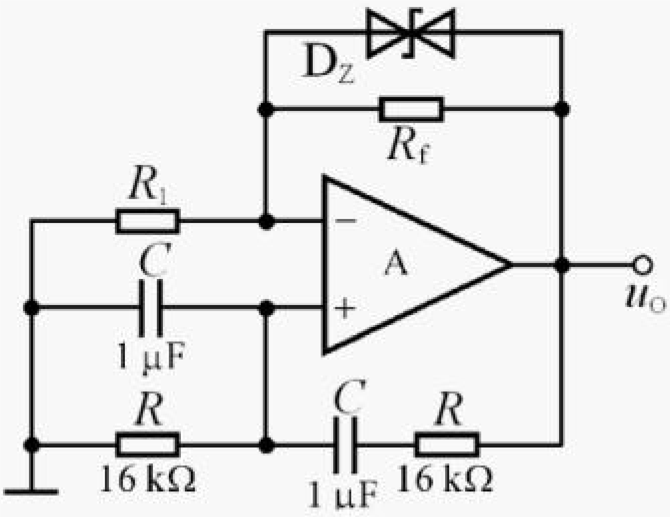

图 P 8.7

8.8 电路如**图 P8.8** 所示。

(1)为使电路产生正弦波振荡，标出集成运放的"+"和"−"；并说明电路是哪种正弦波振荡电路。

(2)若 R_1 短路，则电路将产生什么现象？

(3)若 R_1 断路，则电路将产生什么现象？

(4)若 R_f 短路，则电路将产生什么现象？

(5)若 R_f 断路，则电路将产生什么现象？

8.8 The circuit is shown in figure p8.8.

（1）In order to generate the sine-wave oscillation, mark the "+" "-" of the integrated operational amplifier and judge the type of the oscillation circuit？

（2）If R1 is short circuited, what will happen in the circuit?

（3）If R1 is broken, what will happen in the circuit?

（4）If Rf is short circuited, what will happen in the circuit?

（5）If Rf is broken, what will happen in the circuit?

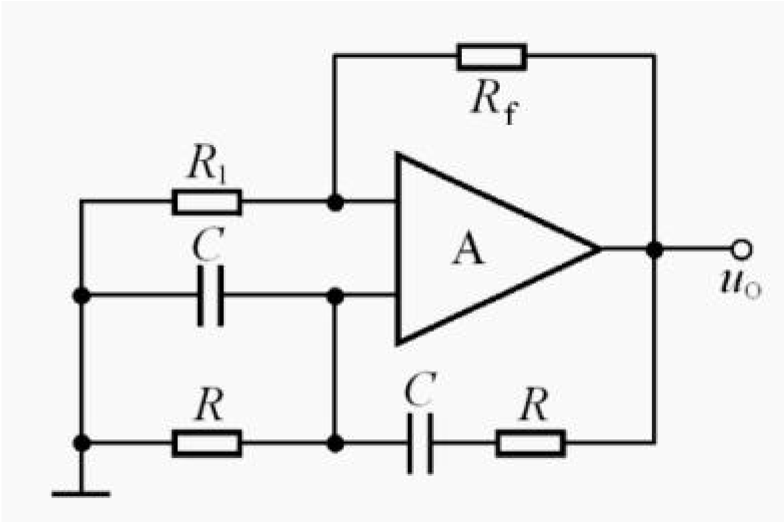

图 P 8.8

8.10 分别标出**图 P8.10** 所示各电路中变压器的同名端，使之满足正弦波振荡的相位条件。

8.10 Please mark the dotted terminal of the mutual inductors in each circuit to generate the sine-wave oscillation as shown in figure p8.10.

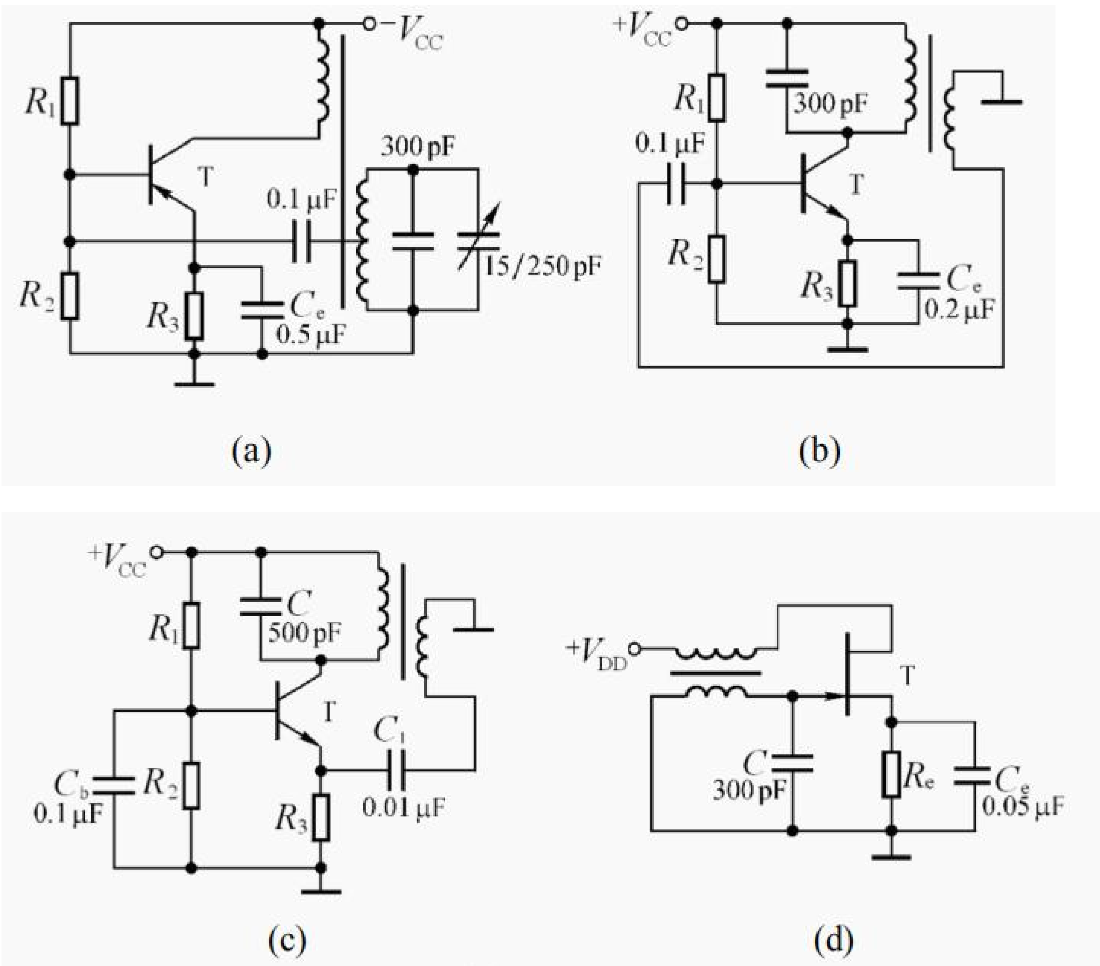

图 P8.10

8.11 分别判断**图 P8.11** 所示各电路是否可能产生正弦波振荡。

8.11 Judge whether the circuits shown in figure p8.11 are able to generate sine-wave oscillation.

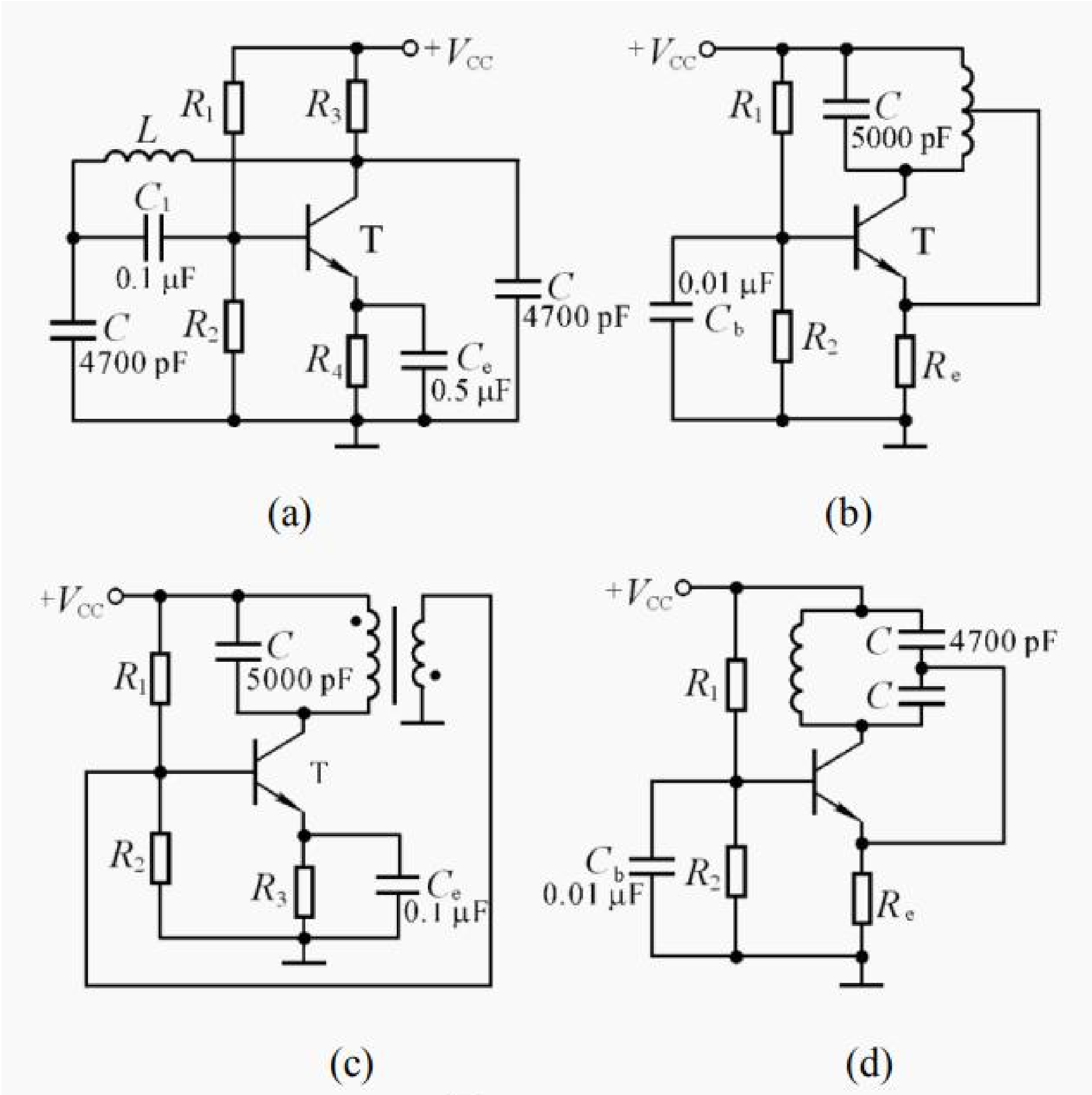

图 P8.11

8.12 改正图 **P8.11**(b)、(c)所示两电路中的错误，使之有可能产生正弦波振荡。

8.12 Correct the errors in the circuits shown in figure p8.11 (b) and figure p8.11 (c) to make it able to generate sine-wave oscillation.